26900047 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBIF303 - UNVEILING MOLECULAR PATTERNS THROUGH CHEMINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **2** marks

1. Identify two applications of cheminformatics in pharmaceutical research. [U] / [CO1]
2. Identify the main use of Open Babel. [U] / [CO6]
3. Write a note about the significance of 2D chemical structure drawing tool. [A] / [CO3, CO4]
4. Determine chemical structure representation. [A] / [CO4]
5. Choose an appropriate descriptor for describing the size of a molecule. [A] / [CO3]
6. Discuss the role of cheminformatics in drug discovery. [U] / [CO6]
7. Discover what type of information is available in PubChem. [A] / [CO2]

[2x5 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

8.	Choose suitable descriptors for predicting the lipophilicity of a series of compounds.	[A] / [CO3]
9.	Describe the factors affecting drug distribution in the body.	[U] / [CO6]
10.	Explain molecular representation.	[A] / [CO3]
11.	Write a note on chemical structure databases.	[A] / [CO2]
12.	Explain how virtual screening is useful in drug discovery.	[A] / [CO5]
13.	Identify and describe molecular descriptors.	[U] / [CO1]
14.	Explain the basic concept of structure-based drug design.	[U] / [CO6]
15.	Judge the importance of molecular descriptors in QSAR studies.	[A] / [CO3]

[5x6 = 30]

Part C

Long Essay

Answer any **3** questions

Each question carries **10** marks

16.	Compare SMILES, InChI and molecular graphs.	[U] / [CO4]
17.	Explain the concept of molecular fingerprints and write its importance in cheminformatics.	[U] / [CO3]
18.	Discuss how polypharmacology differs from network pharmacology.	[U] / [CO6]
19.	Discuss the development of computer-based chemical information systems.	[U] / [CO1]
20.	Give example of cheminformatics tool and explain their uses.	[U] / [CO6]

[10x3 = 30]